

Coding Questions

1. Django Project Setup

Problem:

Write a command to create a new Django project named school_mgmt and start a new app named students.

Input Format:

Project name and App name

Output Format:

Django project and app should be created.

Example 1:

Input:

school_mgmt students

Output:

Project 'school_mgmt' created App

'students' created

Solution :

```
school_mgmt
```

```
django-admin startproject school_mgmt
```

```
cd school_mgmt// changing the directory
```

```
python manage.py startapp students
```

2. Django Model Creation

Problem:

Create a Django model Student with fields: name (CharField), age (IntegerField), email (EmailField).

Input Format:

Model fields

Output Format:

Generated Django model

Example 1 Input:

Student, name:char, age:int, email:email

Output:

```
class Student(models.Model):  
    name = models.CharField(max_length=100) age =  
    models.IntegerField()  
    email = models.EmailField()
```

Solution :

```
from django.db import models  
class Student(models.Model):  
    name = models.CharField(max_length=100)  
    age = models.IntegerField()  
    email = models.EmailField()
```

3. Django URL Mapping

Problem:

Write the urls.py entry to map the URL /students/ to the view student_list.

Input Format:

URL path and view function name

Output Format:

Correct URL mapping in Django

Example 1 Input:

/students/ student_list

Output:

```
path('students/', views.student_list, name='student_list')
```

Solution :

```
from django.urls import path
```

```
from . import views
```

```
urlpatterns = [  
    path('students/', views.student_list, name='student_list'),
```

]

4. Django View and Template

Problem:

Write a view `student_list` that passes a list of student names to the template `students.html`.

Input Format:

List of names

Output Format:

Render template with student names

Example 1 Input:

`["Ali", "Rahul", "Sara"]`

Output:

Template rendered with names: Ali, Rahul, Sara

Solution :

In `views.py`

```
from django.shortcuts import render
```

```
def student_list(request):
```

```
    students = ["Ali", "Rahul", "Sara"]
```

```
    return render(request, 'students.html', {'students': students})
```

`student.html`

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
    <h2>Student List</h2>
```

```
    <ul>
```

```
        {% for student in students %}
```

```
            <li>{{ student }}</li>
```

```
        {% endfor %}
```

```
    </ul>
```

</body>

</html>

5. Django Form Validation

Problem:

Create a Django form for Student with validation ensuring age ≥ 18 .

Input Format:

Name, Age

Output Format:

Form validation message

Example 1 Input:

John, 16

Output:

ValidationError: Age must be at least 18

Example 2 Input:

Alice, 20

Output:

Form is valid

Solution :

```
from django import forms

from django.core.exceptions import ValidationError

class StudentForm(forms.Form):

    name = forms.CharField(max_length=100)

    age = forms.IntegerField()

    def clean_age(self):

        age = self.cleaned_data['age']

        if age < 18:

            raise ValidationError("Age must be at least 18")

        return age
```

6. Django Authentication

Problem:

Write a Django view for user login using `authenticate()` and `login()`.

Input Format:

Username, Password

Output Format:

Login success/failure

Example 1 Input:

(username=admin, password=wrongpass)

Output:

Invalid credentials

Example 2 Input:

(username=admin, password=admin123)

Output:

Login successful

Solution :

```
from django.shortcuts import render, redirect
from django.contrib.auth import authenticate, login
from django.http import HttpResponseRedirect

def user_login(request):
    username = request.POST.get('username')
    password = request.POST.get('password')
    user = authenticate(request, username=username, password=password)
    if user is not None:
        login(request, user)
        return HttpResponseRedirect("Login successful")
    else:
        return HttpResponseRedirect("Invalid credentials")
```

7. MySQL DDL Operation

Problem:

Write an SQL query to create a table courses with id, course_name, credits.

Input Format:

Table name and fields

Output Format:

DDL SQL query

Example Input:

courses id:int, course_name:varchar(100), credits:int

Output:

```
CREATE TABLE courses ( id
    INT PRIMARY KEY,
    course_name VARCHAR(100),
    credits INT
);
```

Solution :

```
CREATE TABLE courses (
    id INT PRIMARY KEY,
    course_name VARCHAR(100),
    credits INT
);
```

8. MySQL DML Operation

Problem:

Write an SQL query to insert a new course record into the courses table.

Input Format:

Course details

Output Format:

SQL Insert statement

Example Input:

(1, "Mathematics", 4)

Output:

```
INSERT INTO courses VALUES (1, 'Mathematics', 4);
```

Solution :

```
INSERT INTO courses VALUES (1, 'Mathematics', 4);
```

9. Flask Routing

Problem:

Create a Flask route `/hello/<name>` that returns "Hello, <name>!".

Input Format:

Name in URL

Output Format:

Greeting string

Example 1 Input:

```
/hello/Ali
```

Output:

```
Hello, Ali!
```

Example 2 Input:

```
/hello/Sara
```

Output:

```
Hello, Sara!
```

Solution :

```
from flask import Flask
app = Flask(__name__)
@app.route('/hello/<name>')
def hello(name):
    return f"Hello, {name}!"
if __name__ == '__main__':
    app.run(debug=True)
```

10. Flask Blueprint

Problem:

Create a Flask blueprint students_bp with a route /students returning "Student List Page".

Input Format:

Blueprint name and route

Output Format:

Blueprint route response

Example 1 Input:

students_bp /students

Output:

Student List Page

Solution :

```
from flask import Blueprint
```

```
students_bp = Blueprint('students', __name__)
```

```
@students_bp.route('/students')
```

```
def students():  
    return "Student List Page"
```

In main app:

```
from flask import Flask
```

```
from students import students_bp
```

```
app = Flask(__name__)
```

```
app.register_blueprint(students_bp)
```

```
if __name__ == '__main__':  
    app.run(debug=True)
```